AJ Motors

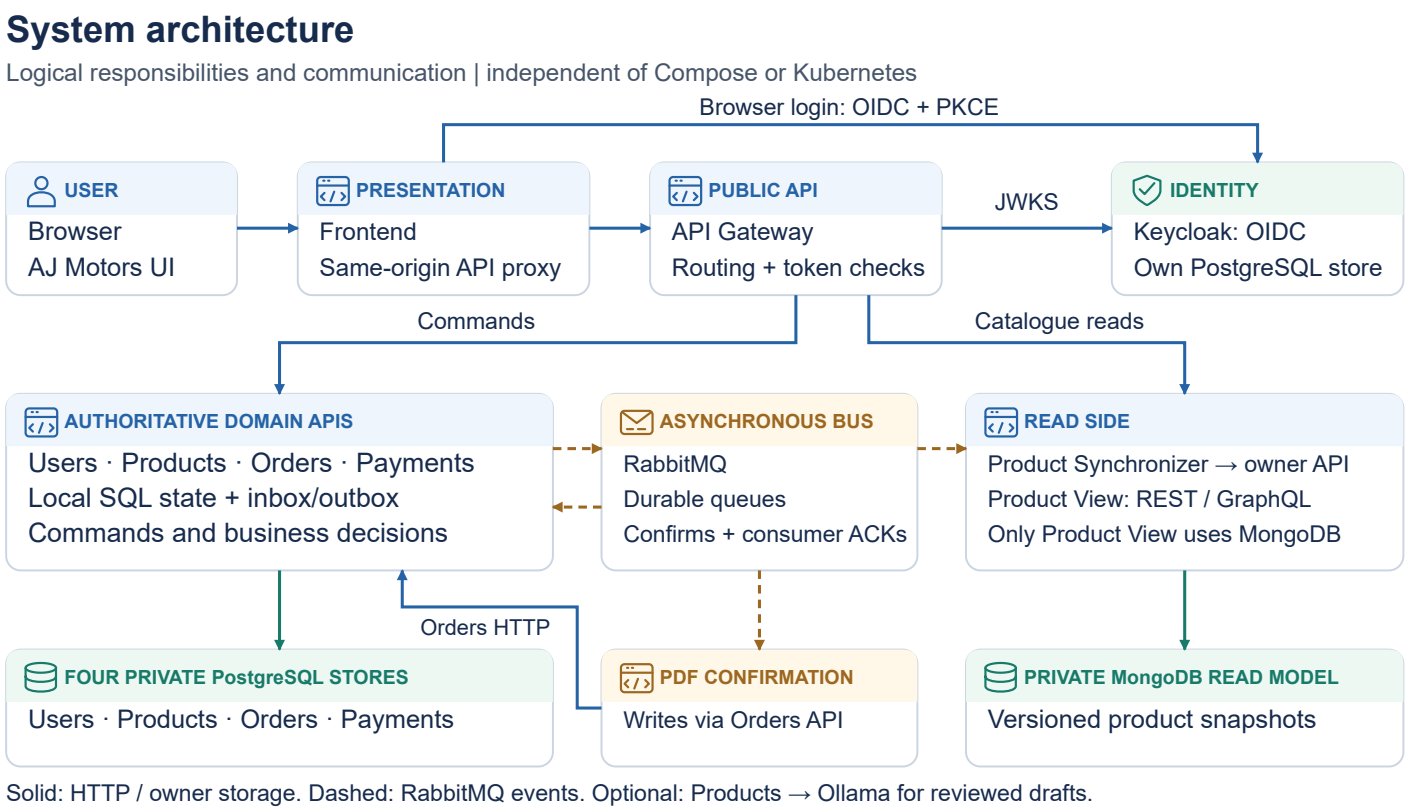
- Distributed car sales · architecture and delivery strategy
- Ahmad Alkaseb · Jeppe Koch · Asger Bergøe · Laith Alkaseb
- https://github.com/AhmadAlkaseb/aj

Problem and acceptance criteria

- Coordinate customer, vehicle, order and manual payment ownership.
- One car must not be reserved by two buyers.
- Frontend consumes GraphQL catalogue queries and REST commands.
- Latency, durability and access boundaries need explicit tests.

System architecture

- Five domain APIs; synchronizer and confirmation processor are additional workers.



Why these technologies?

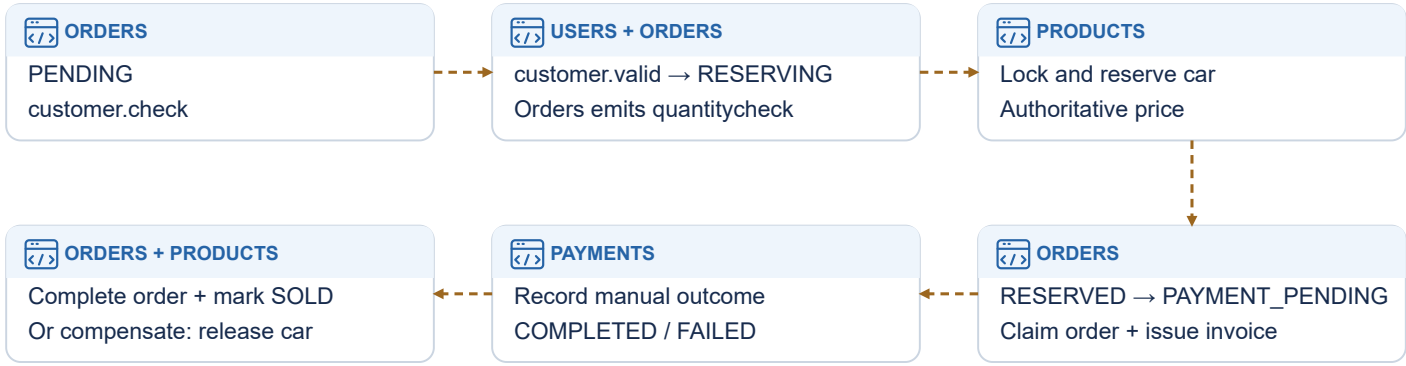
- Java 17/Javalin/Maven: consistent backend and build tooling.
- PostgreSQL: owner constraints and transactions; MongoDB: catalogue snapshots.
- RabbitMQ: asynchronous delivery; REST + GraphQL: commands and queries.
- Compose locally; kind, Calico, HPA and KEDA for the production-like demo.

Communication and patterns

- AMQP coordinates reservation, payment outcomes and catalogue snapshots.
- CQRS, tombstones/snapshots, idempotence and selected commutative handling.
- Saga compensation, deadlines and explicit manual review.
- Outbox/inbox retain work and deduplicate selected redeliveries.

Reservation and payment saga

Each arrow crosses a message boundary; each service commits its own local transaction



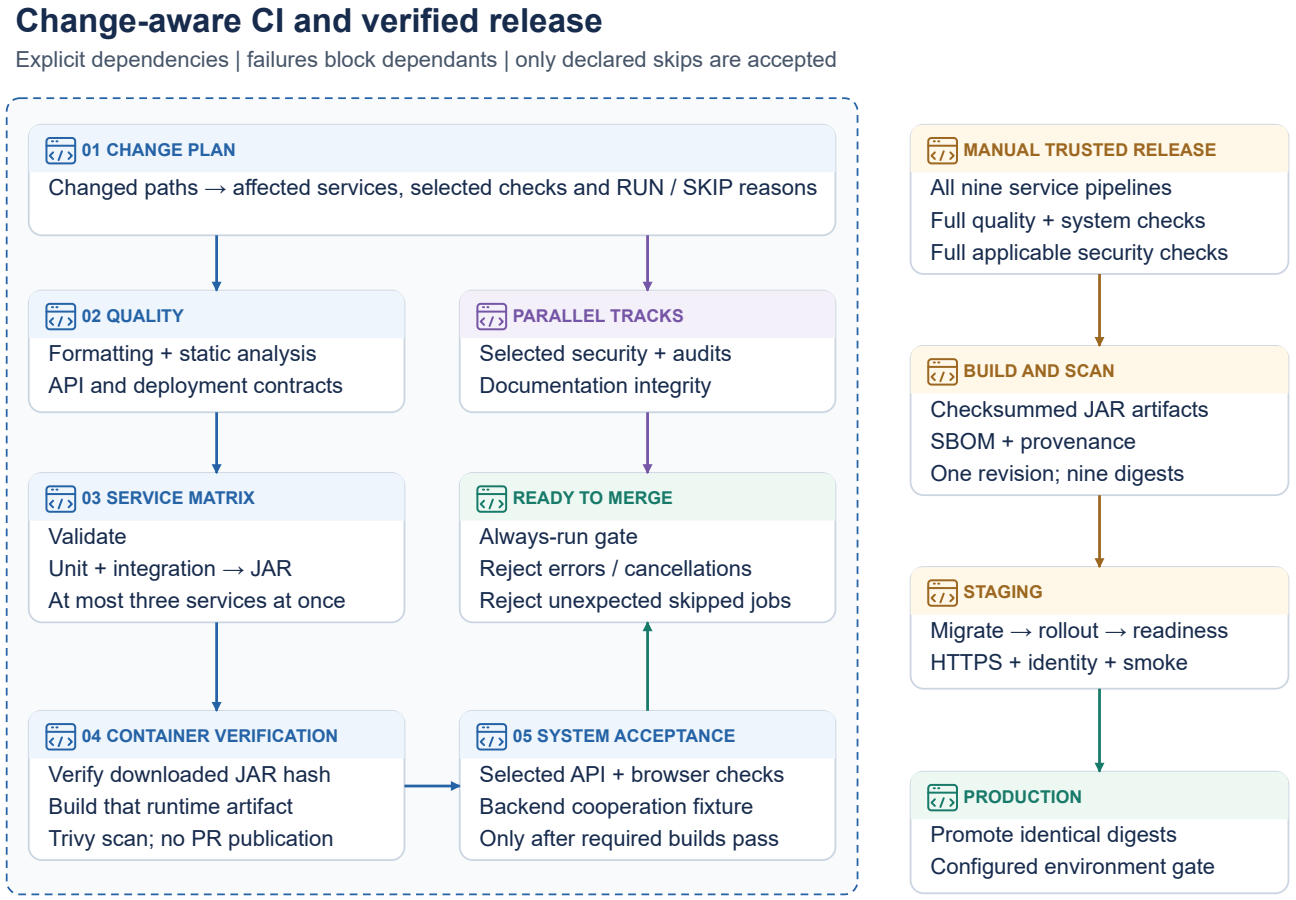
Simplified successful path. Validation rejection, cancellation and deadlines can branch to recovery.

Balanced Scrum-inspired work model

- Four workstreams: identity; catalogue/AI; commerce; operations.
- Each participant owns one stream and reviews another.
- Short planning cycles, shared board, brief standups and acceptance review.
- Allocation/reflection drafts need participant confirmation; no invented attendance.

CI/CD and testing plan

- Change plan → selected quality → affected services → images → cooperation.
- Independent security/docs tracks join one strict Ready to merge gate.
- Unit + integration must pass before JAR packaging.
- Local results and actual GitHub runs remain different evidence.

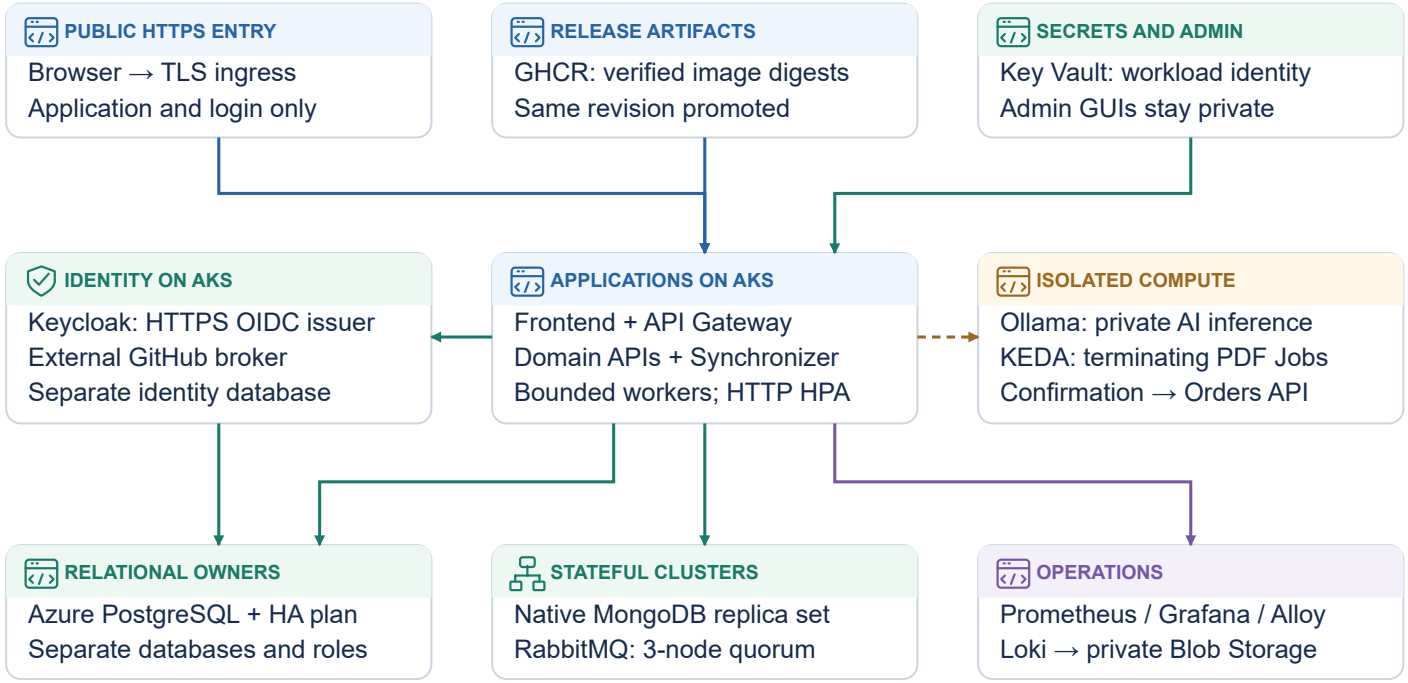


Cloud, operations and versioning

- Selected cloud plan: AKS, PostgreSQL, native MongoDB and RabbitMQ.
- Private identity/AI; KEDA PDF Jobs; secrets and tested recovery.
- Prometheus/Grafana/Alloy/Loki; Git, Flyway, API v1 and event schema.
- Cloud resources are planned, not purchased or deployed.

Proposed Azure deployment

Selected cloud target | not provisioned | private data and operator access



Replicas need separate nodes/zones and tested backups. The local single-node demo does not establish cloud HA.

Delivery and open evidence

- Report with included appendices; monorepo and installation procedure.
- Design and implementation figures are editable repository assets.
- External GitHub login needs an owned OAuth app and real browser verification.
- Upload this PowerPoint, report and repository link to itsLearning.